

An Exploratory Analysis of Participation Patterns in European Citizens Initiatives using Country-Level Socio-Economic Variables

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Background and Research Problem

- The European Citizens' Initiative (ECI) provides EU citizens with a mechanism to propose changes to European policy.
- Participation varies considerably between countries and between individual Initiatives.
- Raw signature counts are strongly influenced by population size.
- Therefore, this dissertation investigates whether country-level socio-economic characteristics can explain participation.
- Participation was examined using:
 - Raw signature counts
 - Signatures per capita

Key problem: Can country-level characteristics explain why citizens participate in ECIs?

RQ1: Raw Signature Counts

Can socio-economic indicators explain raw signature counts?

RQ2: Participation Rates

Can socio-economic indicators explain participation rates?

RQ3: Successful Initiatives

Can socio-economic indicators explain participation rates when the analysis is reduced to only successful Initiatives?

Key comparison:

Complete Dataset vs Successful Initiatives Only

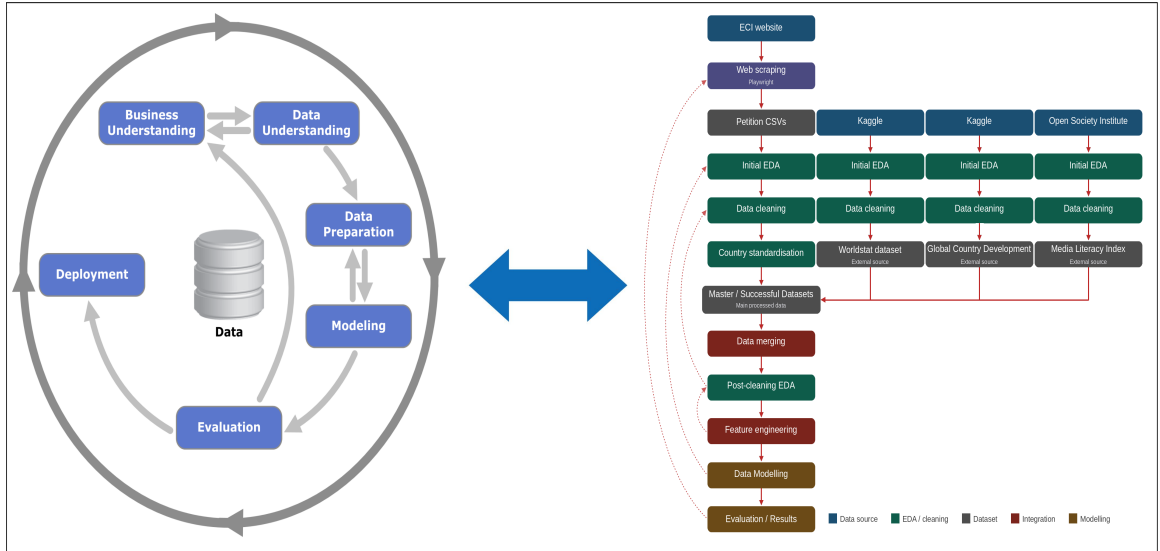
Data

- ECI country-level signature data collected using Python and Playwright.
- Final dataset: 27 Initiatives.
- Combined with external country-level socio-economic datasets.
- Additional indicators included Media Literacy, Governance, Social Capital, Personal Freedom and Education.

Analytical Process

The process followed an iterative CRISP-DM approach, with findings from earlier stages informing later analysis.

Modified CRISP-DM Workflow



Exploratory Data Analysis

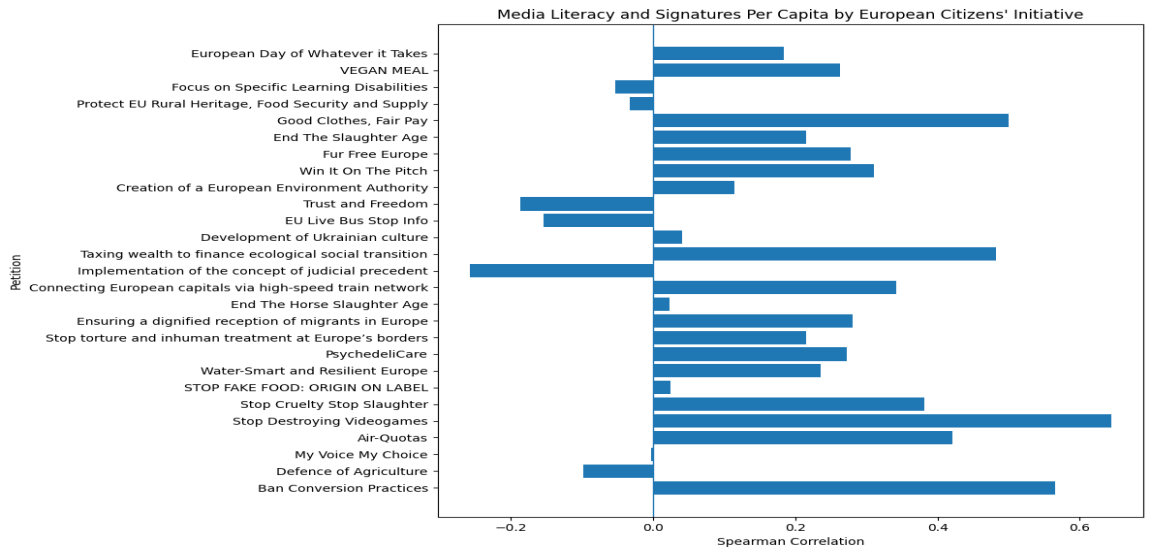
- Population showed a moderate relationship with total signature counts.
- Relationships between many socio-economic indicators and signatures per capita were weak.
- Participation varied substantially between individual Initiatives.
- ANOVA conducted during the interim stage provided strong evidence that participation differed between Initiatives.
- Media Literacy Index analysis also showed that relationships varied considerably between individual Initiatives.

Key Finding

Participation could not be treated as homogeneous across all ECIs.

This finding motivated the creation of a second dataset containing only the successful Initiatives.

Media Literacy vs Signatures Per Capita by Petition



Research Question 1 - Raw Signature Counts

Can socio-economic indicators explain raw signature counts?

Three linear regression models were developed:

- **1A:** Population + GDP Per Capita
- **1B:** Population + Density + Tertiary Enrollment + Life Expectancy
- **1C:** Population + Media Literacy + Governance + Social Capital + Personal Freedom + Education

	1A	1B	1C
R^2	0.0999	0.0994	0.1013
Adjusted R^2	0.0974	0.0944	0.0939

Finding: $\approx 10\%$ of variation explained.

Interpretation

Population size was the main driver, rather than broader socio-economic characteristics.

Research Question 2 - Participation Rates

Can socio-economic indicators explain participation rates?

Participation was normalised using:

$$\text{Signatures Per Capita} = \frac{\text{Signatures}}{\text{Population}}$$

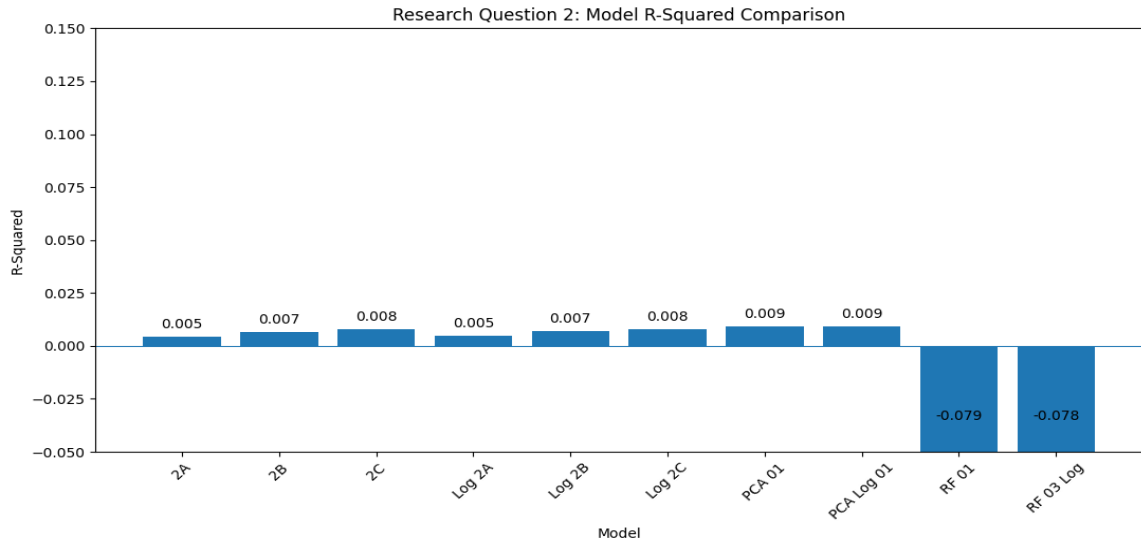
Results across modelling approaches:

Model	R ²
Linear 2C	0.0079
Log Linear 2C	0.0080
PCA Regression 01	0.0089
PCA Log Regression 01	0.0089

Finding

Socio-economic variables provided very limited explanatory across the complete dataset.

Question 2 Model R-Squared Values



Research Question 3 - Successful Initiatives

Can socio-economic indicators explain participation rates for successful Initiatives?

The dataset was reduced from 27 Initiatives to the **4 successful Initiatives**.

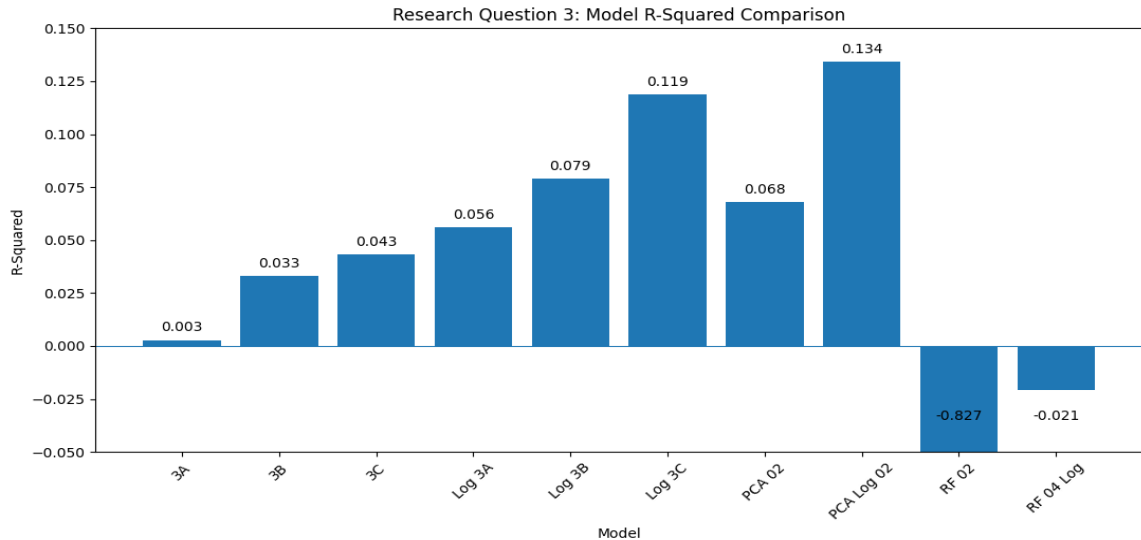
Model	R^2	Model p-value
Linear 3C	0.0431	0.4721
Log Linear 3C	0.1187	0.0227
PCA Regression 02	0.0679	0.1201
PCA Log Regression 02	0.1341	0.0048

Best result: $R^2 = 0.1341$ using the logarithmic PCA regression.

Interpretation

Significant improvement was observed, but the small successful dataset limits the reliability of this finding.

Question 3 Model R-Squared Values



Cluster Analysis and Overall Findings

Cluster Analysis

- K-means clustering was used to group countries based on socio-economic variables.
- The elbow method was used to select the number of clusters.
- Four clusters were selected.
- Clusters were visualised geographically across Europe.

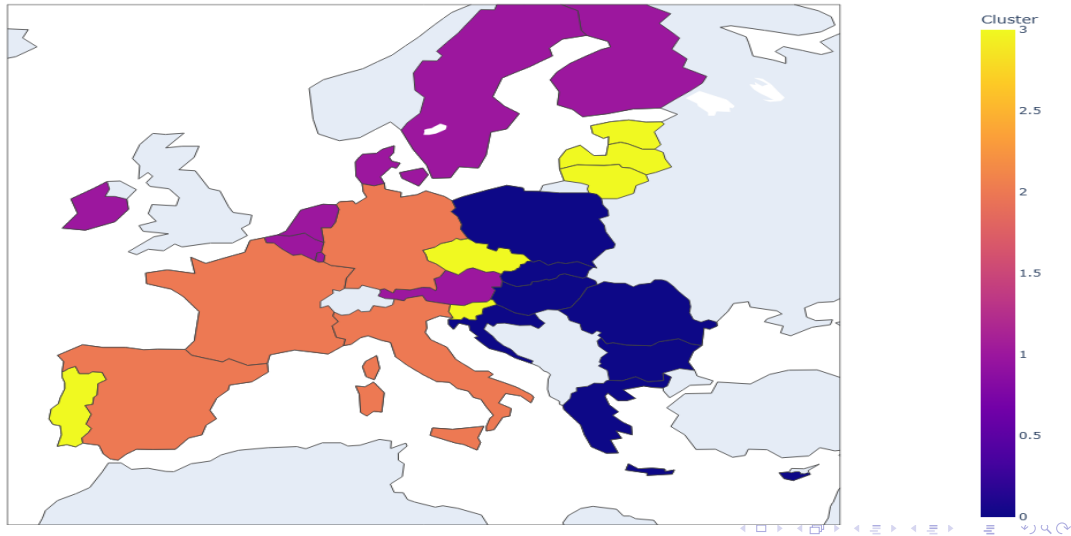
Overall Finding

Country-level socio-economic characteristics alone cannot adequately explain ECI participation.

- Raw signatures were strongly influenced by population.
- Participation rates were much harder to explain.
- Successful Initiatives showed some improvement.
- Initiative-specific factors appear more important.

European Countries by Cluster

European Countries by Cluster



Conclusion and Future Work

Conclusions

- RQ1: Socio-economic variables explained approximately 10% of raw signature variation, largely influenced by population.
- RQ2: Very little evidence that country-level socio-economic variables explain participation rates across all Initiatives.
- RQ3: Restricting the analysis to successful Initiatives produced stronger results.
- Random Forest models struggled to generalise and showed evidence of overfitting.

Future Work

- Increase the number of Initiatives and observations.
- Incorporate Initiative-level and campaign-level variables.
- Analyse issue and campaign characteristics.
- Explore social media and Natural Language Processing approaches.

ECI participation is a multi-factor problem beyond country-level socio-economic factors.

Any Questions?

Thank you for listening.